

CE 512: Advanced Soil Mechanics

Course Code	CE 512
Course Name	Advanced Soil Mechanics
L-T-P-C	3-0-0-3
Prerequisites	CE302 or Equivalent
Intended for	UG and PG
Distribution	Discipline Elective
Approval	36th BoA (Sep 2020); 28th Senate (Nov 2020), in the B.Tech Civil Engineering curriculum

Course Contents

- **Steady State Flow through Soil**
Laplace's Equation of Continuity; Permeability; Flow net, use and method of obtaining flow net; Numerical solution for 2D steady state flow in soil; Examples of 2D and 3D seepage in soil, seepage in anisotropic soil. [8 Hours]
- **Transient Flow in Soil**
Compressibility and rate of consolidation; Primary, secondary and tertiary consolidation; Consolidation theories: Terzaghi and Biot's formulation, numerical solution of governing consolidation equation; Laboratory consolidation test and its interpretation. [8 Lectures]
- **Strength and Deformation Behavior of Soil**
Introduction to stress-strain behavior of soils, principal stresses, Mohr diagrams; Shear strength of cohesive and cohesionless soils, various failure criteria, drained and undrained shear strength of soils; Determination of shear strength from various laboratory tests, e.g. direct shear, triaxial, simple shear, true triaxial, hollow cylinder test, interpretation of various triaxial test results, significance of pore pressure parameters; Concept of critical void ratio; Dilation in sands. [11 Lectures]
- **Stress Path Pertaining to Various Loading Conditions**
Stress path, drained and undrained stress path; Stress path with respect to different initial state of the soil; Stress path for different practical situations. [4 Lectures]
- **Critical State Concept**
Critical state soil mechanics, concept of soil yielding, critical state parameters; Critical state for normally consolidated and over consolidated soil; Significance of Roscoe and Hvorslev state boundary surface, drained and undrained plane, complete state boundary surface. [11 Lectures]

Text Books

1. Muni Budhu, **Soil Mechanics and Foundations**, John Wiley & Sons, 2010.
2. T.W. Lambe and R.V. Whitman, **Soil Mechanics**, John Wiley & Sons, 1991

References

1. Jonathan Knappett and R.F. Craig, **Craig's Soil Mechanics**, CRC Press, 2012.
2. B.M. Das, **Advanced Soil Mechanics**, CRC Press, 2013.
3. James K. Mitchell, Kenichi Soga, **Fundamentals of Soil Behavior**, John Wiley & Sons, 2005.
4. J.H. Atkinson, **The Mechanics of Soils and Foundations**, CRC Press, 2007.
5. Holtz, R. D., and Kovacs, W. D., **An Introduction of Geotechnical Engineering**, Prentice Hall, 1981.
6. Parry, R. H. G., **Mohr Circles, Stress Paths and Geotechnics**, CRC Press, 2004.
7. David Muir Wood, **Soil Behaviour and Critical State Soil Mechanics**, Cambridge University Press, 1991.
8. Andrew Schofield and Peter Wroth, **Critical State Soil Mechanics**, McGraw Hill, 1968.
9. Potts, D.M. and Zdravkovic, L., **Finite Element Analysis in Geotechnical Engineering: Theory**, Thomas Telford, USA, 1999.